



MLLaB

=====Kmeans=====

```
# -----Kmeans-----  
from sklearn import datasets  
from sklearn import metrics  
from sklearn.cluster import KMeans  
from sklearn.model_selection import train_test_split  
iris = datasets.load_iris()  
X_train,X_test,y_train,y_test = train_test_split(iris.data,iris.target)  
model =KMeans(n_clusters=3)  
model.fit(X_train,y_train)  
kmean=metrics.accuracy_score(y_test,model.predict(X_test))  
print("acuracy of Kmean",kmean)
```

#-----Expectation and Maximization-----

```
from sklearn.mixture import GaussianMixture  
model2 = GaussianMixture(n_components=5)  
model2.fit(X_train,y_train)  
model2.score  
em=metrics.accuracy_score(y_test,model2.predict(X_test))  
print("acuracy of EM",em)
```

=====Weighted Regression=====

```
import numpy as np  
import matplotlib.pyplot as plt  
x = np.linspace(-5, 5, 1000)  
y = np.log(np.abs((x ** 2) - 1) + 0.5)  
#y=np.sin(x)
```

```
x = x + np.random.normal(scale=0.05, size=1000)
plt.scatter(x, y, alpha=0.3)
```

```
def local_regression(x0, x, y, tau):
    x0 = np.r_[1, x0]
    x = np.c_[np.ones(len(x)), x]
    xw = x.T * radial_kernel(x0, x, tau)
    beta = np.linalg.pinv(xw @ x) @ xw @ y
    return x0 @ beta

def radial_kernel(x0, x, tau):
    return np.exp(np.sum((x - x0) ** 2, axis=1) / (-2 * tau ** 2))

def plot_lr(tau):
    domain = np.linspace(-5, 5, num=300)
    pred = [local_regression(x0, x, y, tau) for x0 in domain]
    plt.scatter(x, y, alpha=0.3)
    plt.plot(domain, pred, color="red")
    return plt

plot_lr(0.03).show()
```

===== k-Nearest Neighbour=====

-----Kmeans-----

```
from sklearn import datasets
from sklearn import metrics
from sklearn.cluster import KMeans
from sklearn.model_selection import train_test_split
iris = datasets.load_iris()
X_train,X_test,y_train,y_test = train_test_split(iris.data,iris.target)
model =KMeans(n_clusters=3)
model.fit(X_train,y_train)
kmean=metrics.accuracy_score(y_test,model.predict(X_test))
print("acuracy of Kmean",kmean)
```

#-----Expectation and Maximization-----

```
from sklearn.mixture import GaussianMixture
model2 = GaussianMixture(n_components=5)
```

```
model2.fit(X_train,y_train)
model2.score
em=metrics.accuracy_score(y_test,model2.predict(X_test))
print("accuracy of EM",em)
```

//Dataset:

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